

CSE 4417 IoT Theory

Lecture 04: Actuation & Actuators

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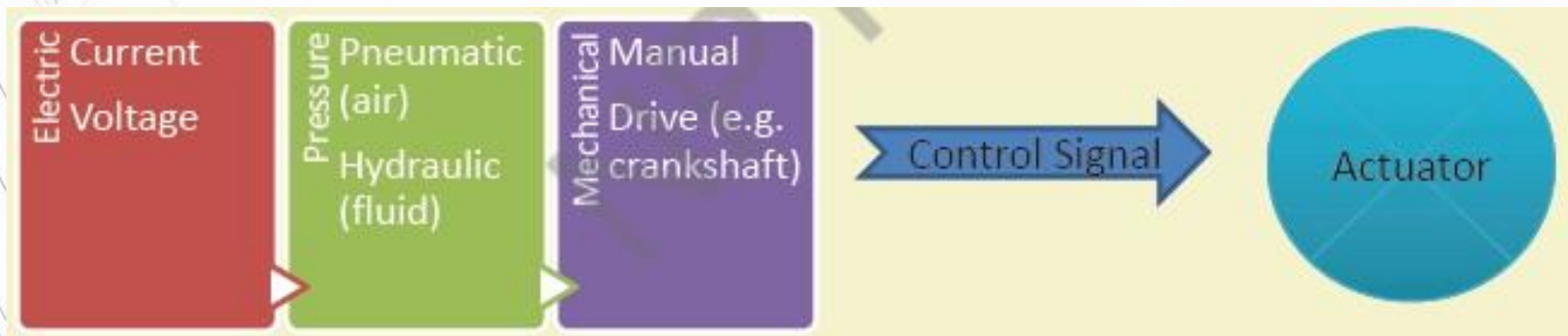
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Actuator

- An actuator is a component of a machine or system that moves or controls the mechanism or the system.
- An actuator is the mechanism by which a control system acts upon an environment
- An actuator requires a control signal and a source of energy.

Actuator

- Upon receiving a control signal is received, the actuator responds by converting the energy into mechanical motion.
- The control system can be simple (a fixed mechanical or electronic system), software-based (e.g. a printer driver, robot control system), a human, or any other input.



Actuator type



Hydraulic Actuators

- A hydraulic actuator consists of a cylinder or fluid motor that uses hydraulic power to facilitate mechanical operation.
- The mechanical motion is converted to linear, rotary or oscillatory motion.
- Since liquids are nearly impossible to compress, a hydraulic actuator exerts considerable force.
- The actuator's limited acceleration restricts its usage.

Hydraulic Actuators



P&H Series
Pressure Switch



Pneumatic Actuators

- A pneumatic actuator converts energy formed by vacuum or compressed air at high pressure into either linear or rotary motion.
- Pneumatic rack and pinion actuators are used for valve controls of water pipes.
- Pneumatic energy quickly responds to starting and stopping signals.

Pneumatic Actuators

- Pneumatic actuators enable large forces to be produced from relatively small pressure changes (e.g., Pneumatic brakes can be very responsive to small changes in pressure applied by the driver)
- It is responsible for converting pressure into force

Pneumatic Actuators



Electric Actuators

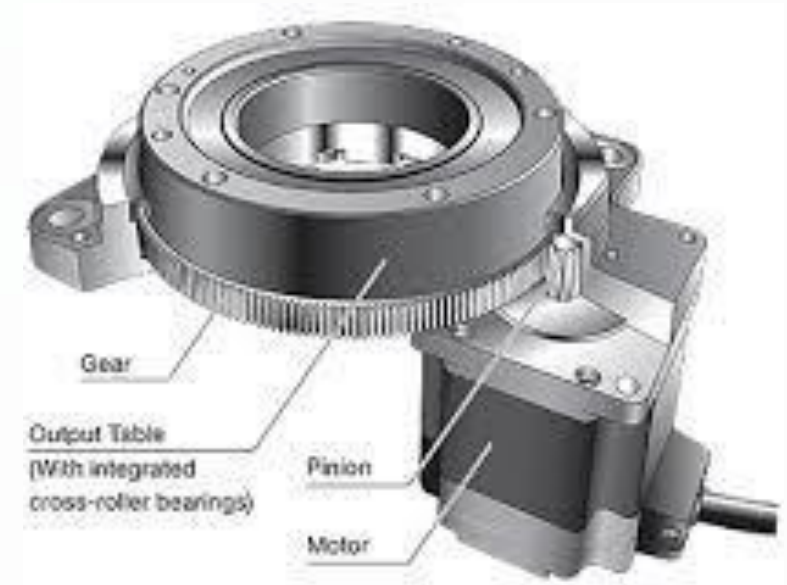
- An electric actuator is generally powered by a motor that converts electrical energy into mechanical torque.
- The electrical energy is used to actuate equipment such as solenoid valves which control the flow of water in pipes in response to electrical signals.
- Considered as one of the cheapest, cleanest and speedy actuator types available.

Electric Actuators

The clapper swings back!



A solenoid based electric bell ringing mechanism

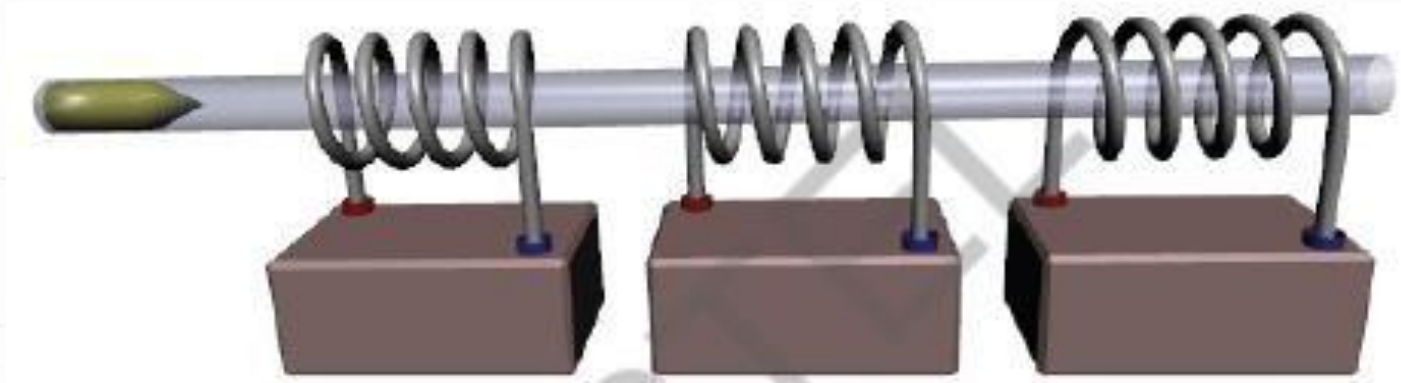


A motor drive-based rotary actuator

Thermal or Magnetic Actuators

- These can be actuated by applying thermal or magnetic energy.
- They tend to be compact, lightweight, economical and with high power density.
- Some popular manufacturers of these devices are *Finnish Modti Inc.* and *American Dynalloy*.

Thermal or Magnetic Actuators

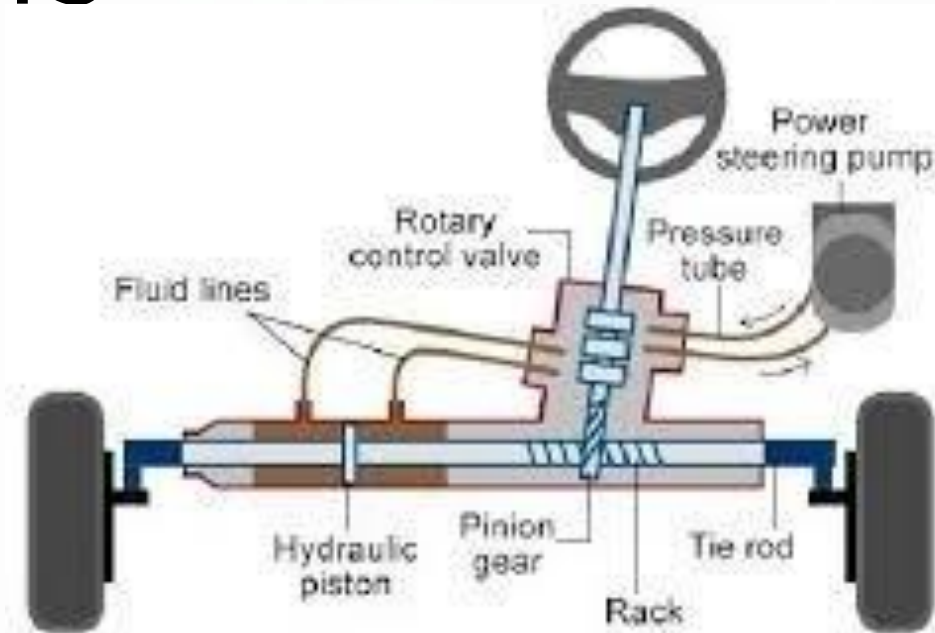
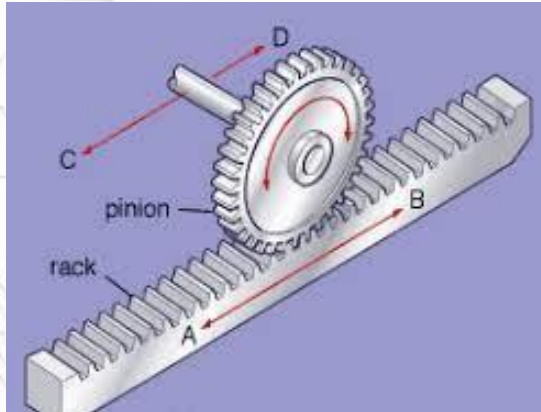


A coil gun works on the principle of magnetic actuation

Mechanical Actuators

- A mechanical actuator converts rotary motion into linear motion to execute some movement.
- It involves gears, rails, pulleys, chains and other devices to operate.
- Example: rack and pinion.

Mechanical Actuators



A rack and pinion mechanism

MECHANICAL



VERSUS

MAGNETIC



Soft Actuators

- Soft actuators (e.g. polymer based) are designed to handle fragile objects like fruit harvesting in agriculture or manipulating the internal organs in biomedicine.
- They typically address challenging tasks in robotics.
- Soft actuators produce flexible motion due to the integration of microscopic changes at the molecular level into a macroscopic deformation of the actuator materials

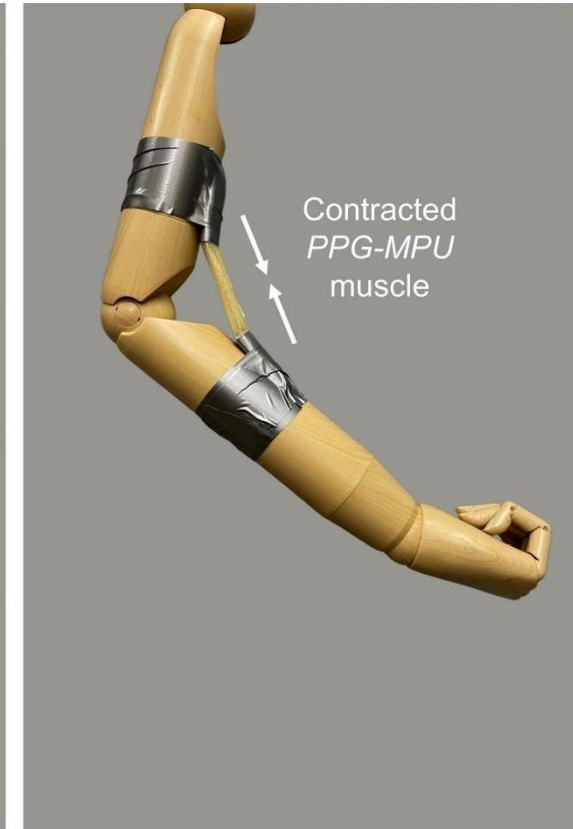
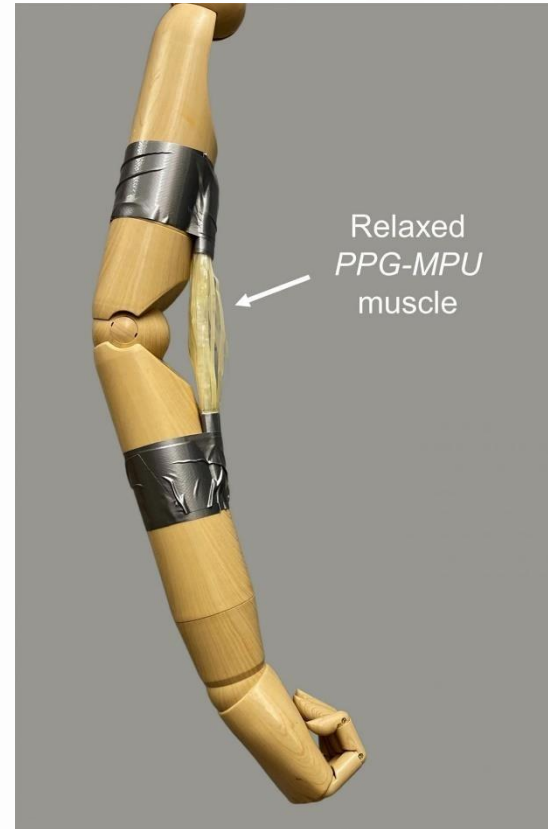
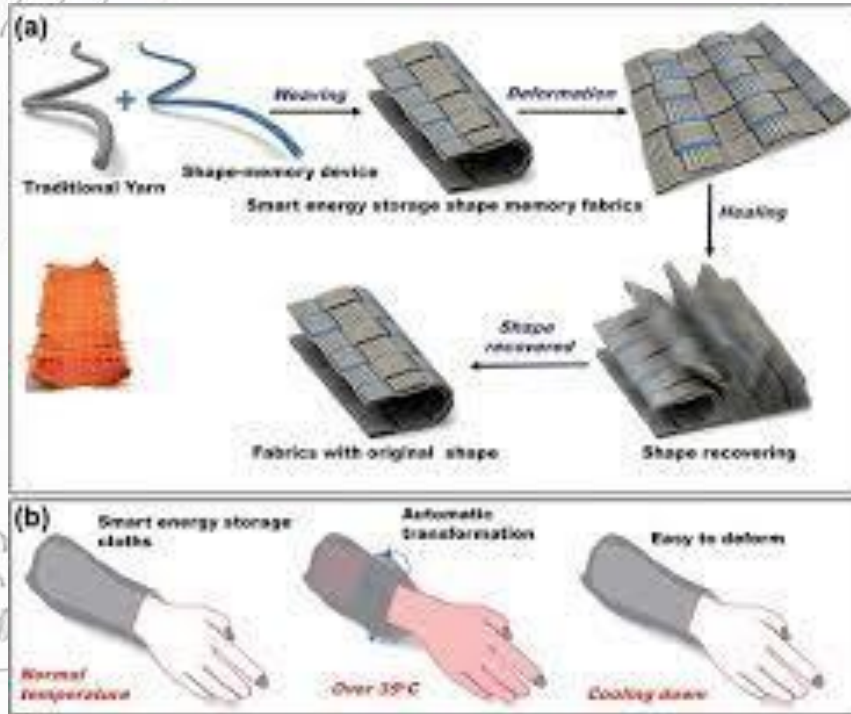
Soft Actuators



Shape Memory Polymers

- Shape memory polymer (SMP) actuators function similar to our muscles, even providing a response to a range of stimuli such as light, electrical, magnetic, heat, pH, and moisture changes.
- SMP exhibits surprising features such a low density, high strain recovery, biocompatibility, and biodegradability

Shape Memory Polymers





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Thank You

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